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Modes of Operation

- → ECB (Electronic codebook)
- → CBC (Cipher block chaining)
- → CFB (Cipher feedback)
- → OFB (Output Feedback)
- → CRT (Counter)

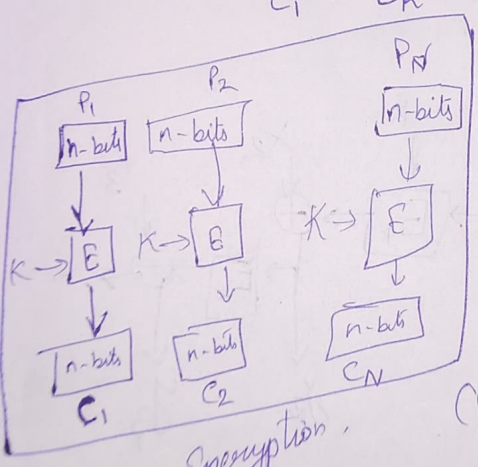
Electronic codebook :

Block Cipher : 64-bit.

↳ Independent way (Parallel Encryption Allowed)

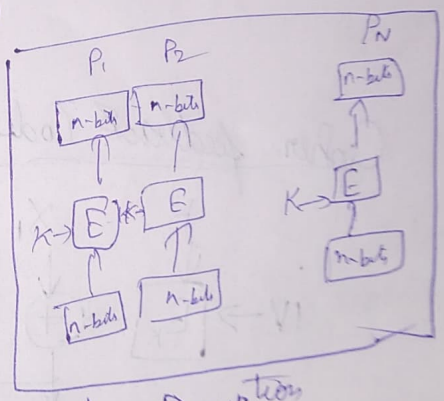
(Key will be the same secret key)

$$c_i = E_K(P_i)$$



Encryption.

(Parallel)



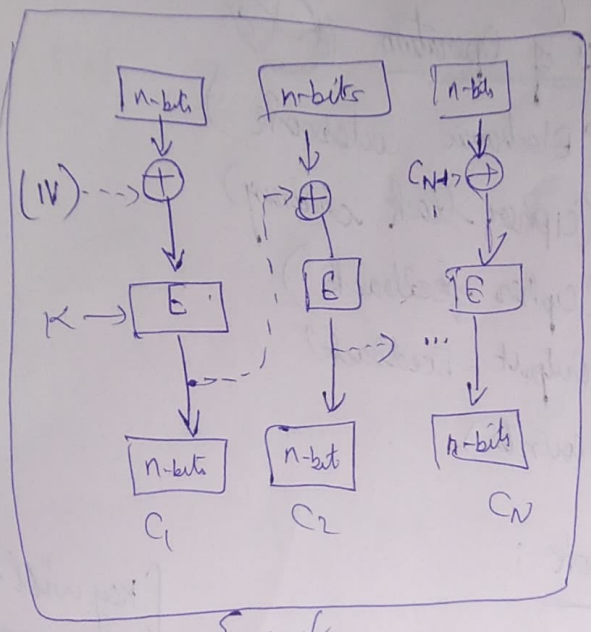
Decryption

This is not secure because for same plain text we get same encryption.

To overcome we use (IV) Initial Value

Cipher Block chaining

Not a parallel Process.



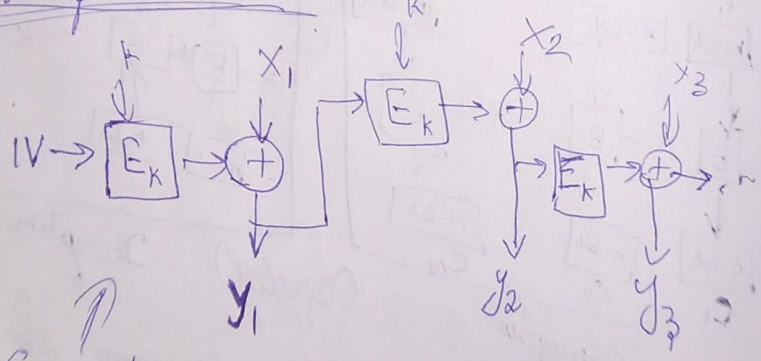
Encryption

C_0 = Initial Vector (IV)

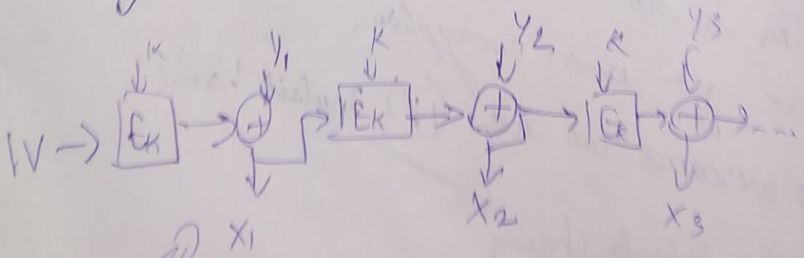
$C_i = E_K(P_i \text{ XOR } C_{i-1})$

(It is a secure Process)

Cipher feedback Code:



Encryption



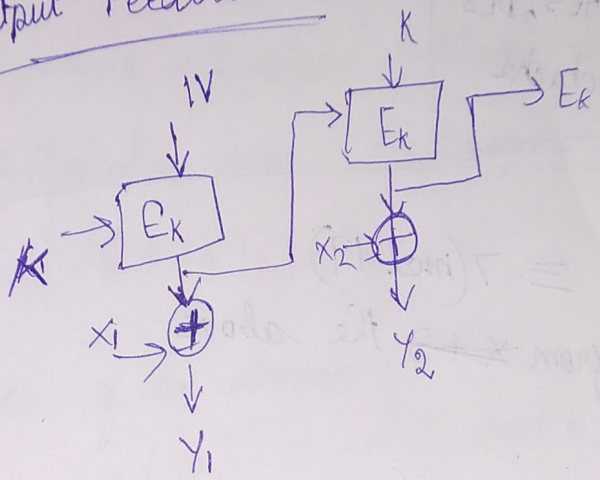
Decryption

8 bits ⇒ [unit]

If the data is < 8 bit use padding method.

Pseudo random no: Pattern may repeat to padd the data...

Output Feedback Mode



$$C_i = P_i \oplus O_i$$

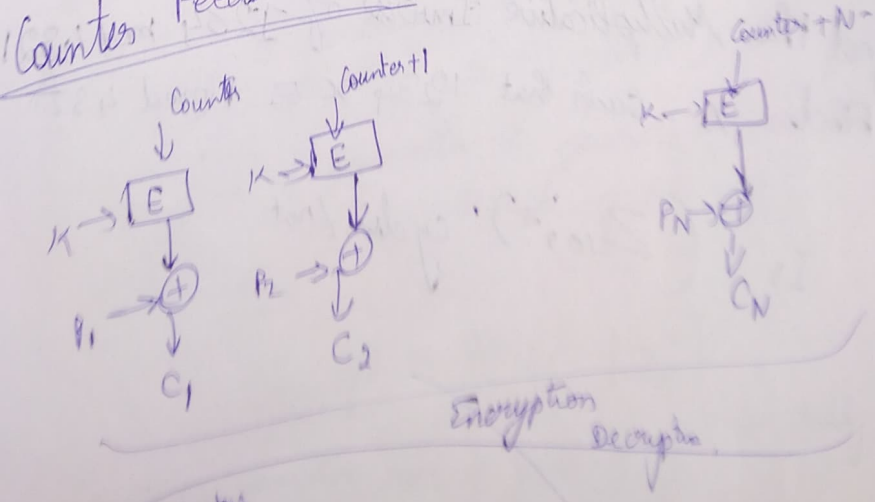
$$O_i = E_K(O_{i-1})$$

$$O_0 = IV$$

Number used once

Counter Feedback Mode (CTR)

[Key is calculated using Nonce]



Keystream $i = E(K, CTR\,i)$

$C_i = P_i \oplus \text{keystream}_i$

Block Cipher : AES, DES
Stream Cipher: chacha

Find $7x + 3 \equiv 7 \pmod{19}$

Find X from ~~x+3~~ the above

$7x + 3 \equiv 7 \pmod{19}$

$7x \equiv 7 - 3 \pmod{19}$

$7x \equiv 4 \pmod{19}$

$x = 6$

1) Find the Multiplicative Inverse of 1234 mod 4321

(a) Find x such that $1234\,x \equiv 1 \pmod{432}$

Is $(Z_{10}, +)$ cyclic/not

Find the Primitive Rule

Euclidean Algorithm